

Lab3- Network Security

Part 1:

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
6416	181.473555	2607:f8b0:4023:1002::	2600:100c:a221:d14a::	TCP	86	[TCP Keep-Alive ACK] 5228 → 53786 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
6417	181.635528	2600:100c:a221:d14a::	2a06:98c1:3100::681::	TCP	75	[TCP Keep-Alive] 61388 → 443 [ACK] Seq=0 Ack=1 Win=253 Len=1
6418	182.492140	2a06:98c1:3100::681::	2600:100c:a221:d14a::	TCP	86	[TCP Keep-Alive ACK] 443 → 61388 [ACK] Seq=1 Ack=1 Win=17 Len=0 SLE=0 SRE=1
6419	183.381988	192.168.1.183	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
6420	183.382669	fe80::9df2:ee02:3ee::ff02::fb	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
6421	183.382669	fe80::9df2:ee02:3ee::ff02::fb	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
6422	183.382669	fe80::9df2:ee02:3ee::ff02::fb	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
6423	184.187660	2600:100c:a221:d14a::	2607:f8b0:4023:1002::	TCP	75	[TCP Keep-Alive] 59638 → 5228 [ACK] Seq=1 Ack=1 Win=252 Len=1
6424	184.189642	192.168.1.183	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
6425	184.645606	2607:f8b0:4023:1002::	2600:100c:a221:d14a::	TCP	86	[TCP Keep-Alive ACK] 5228 → 59638 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
6426	184.645710	WNC_72:09:00	CloudNetwork_7b:3c::	ARP	42	Who has 192.168.1.202? Tell 192.168.1.1
6427	184.645739	CloudNetwork_7b:3c::	WNC_72:09:00	ARP	42	192.168.1.202 is at 10:b1:df:7b:3c:bb

> Frame 1: Packet, 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface \Device\NPF_{16335CB7-C7-41-FE-72-09-00} (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2a06:98c1:3100::ac40:94eb

> Transmission Control Protocol, Src Port: 59245, Dst Port: 443, Seq: 1, Ack: 1, Len: 90

> Transport Layer Security

Wi-Fi <live capture in progress> | Packets: 6427 | Profile: Default

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
87	28.152657	fe80::41b:5ccf:7e06::ff02::fb	ff02::fb	MDNS	271	Standard query response 0x0000 TXT, cache flush NSEC, cache flush iPad (2)...companion-link._tcp.1
88	29.174066	2a06:98c1:3100::ac40::	2600:100c:a221:d14a::	TCP	86	[TCP Dup ACK #41] 443 → 59245 [ACK] Seq=49 Ack=57 Win=17 Len=0 SLE=29 SRE=57
89	29.246332	192.168.1.202	192.168.1.1	DNS	79	Standard query 0x6dc8 A default.exp-tas.com
90	29.246592	192.168.1.202	192.168.1.1	DNS	79	Standard query 0x1f5e AAAA default.exp-tas.com
91	29.250592	192.168.1.1	192.168.1.202	DNS	164	Standard query response 0x1f5e AAAA default.exp-tas.com CNAME deault-exp-tas-com.e-0014.e-msedge.
92	29.898044	192.168.1.1	192.168.1.202	DNS	161	Standard query response 0x6dc8 A default.exp-tas.com CNAME deault-exp-tas-com.e-0014.e-msedge.net
93	29.925405	192.168.1.202	13.107.5.93	TCP	66	51406 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
94	30.054060	2600:100c:a221:d14a::	2a06:98c1:3100::681::	TCP	75	61816 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
95	30.057100	13.107.5.93	192.168.1.202	TCP	66	443 → 51406 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
96	30.057235	192.168.1.202	13.107.5.93	TCP	54	51406 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
97	30.115013	192.168.1.202	13.107.5.93	TLSv1	500	Client Hello (SNI=default.exp-tas.com)
98	30.514716	192.168.1.202	13.107.5.93	TCP	500	[TCP Retransmission] 51406 → 443 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=446

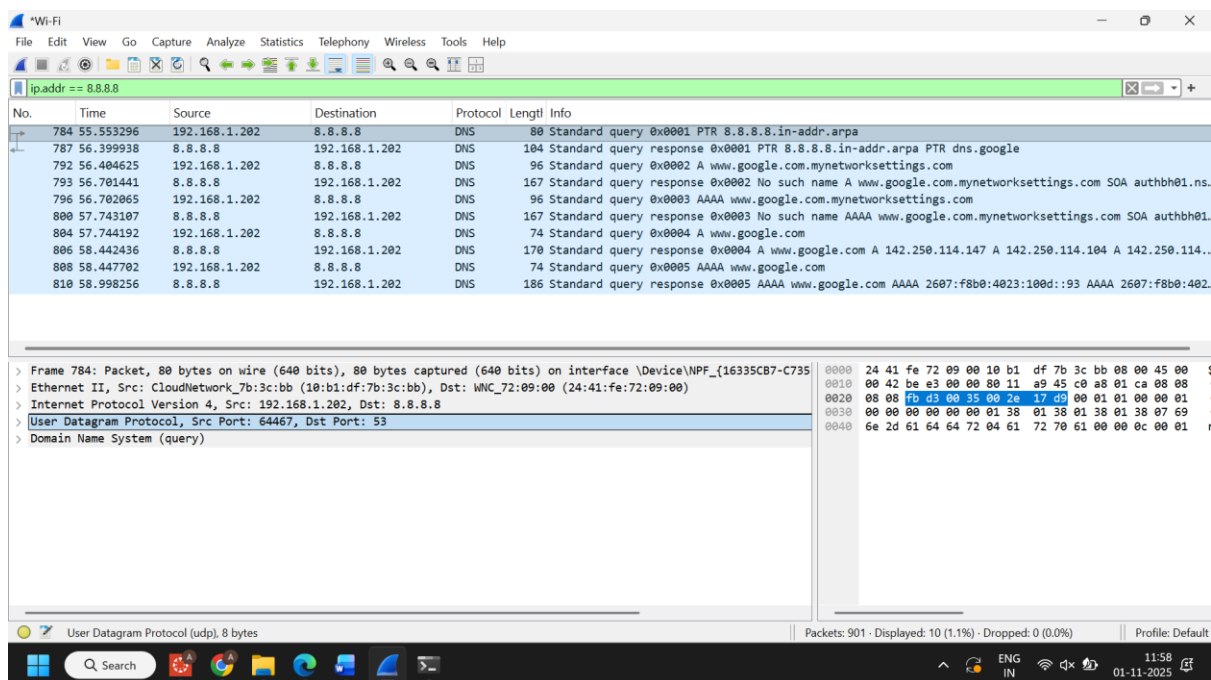
> Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{16335CB7-C7-41-FE-72-09-00} (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2603:1063:27:1::14

> Transmission Control Protocol, Src Port: 60793, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

Wi-Fi <live capture in progress> | Packets: 98 | Profile: Default



Q1. The Length of each and every UDP header field.

Field Size

Source Port 2 bytes

Destination Port 2 bytes

Length 2 bytes

Checksum 2 bytes

Total UDP header = 8 bytes

Q2.

Payload = Length - 8.

From your capture:

UDP Length = 80 bytes (see Frame 784).

So payload = 80 - 8 = 72 bytes.

header and data Length and payload is 72 bytes.

Q3. Maximum UDP payload size

UDP Length field: is 16bit - maximum = 65 535 bytes.

Subtract 8-byte header:

65 535 [?] 8 = 65 527 bytes

Here The maximum UDP packet size is = 65 527 bytes.

Q4. Rank of largest possible source port.

Port fields are 16 bits - range 0 - 65 535.

Answer: 65 535

Q5. the UDP protocol number is:

In the IP header - Protocol = UDP (17).

Decimal = 17 Hex = 0x11

Q6. Request and reply port number correlation.

Source IP Port Direction IP Port Destination IP Port

Request 192.168.1.202 : 64467 8.8.8.8 : 53

Reply 8.8.8.8 : 53 192.168.1.202 : 64467

Response: The request and the response port are exchanged (between the client and the server): the

The destination port (server response) is request port (temporary) of the client.

Other Activities: Performance Analysis: UDP:

Q7. UDP: faster than TCP Why / When to use it.

Speed reasons

No handshake (less delay)

No notifications and reservations.

Decreased length of the header is (8 bytes versus 20 in TCP).

Use cases

DNS queries (short and quick)

Streaming (audio/video)

VoIP and online gaming (live information).

UDP is faster because it is connectionless and lightweight, it is executed in time sensitive mode.

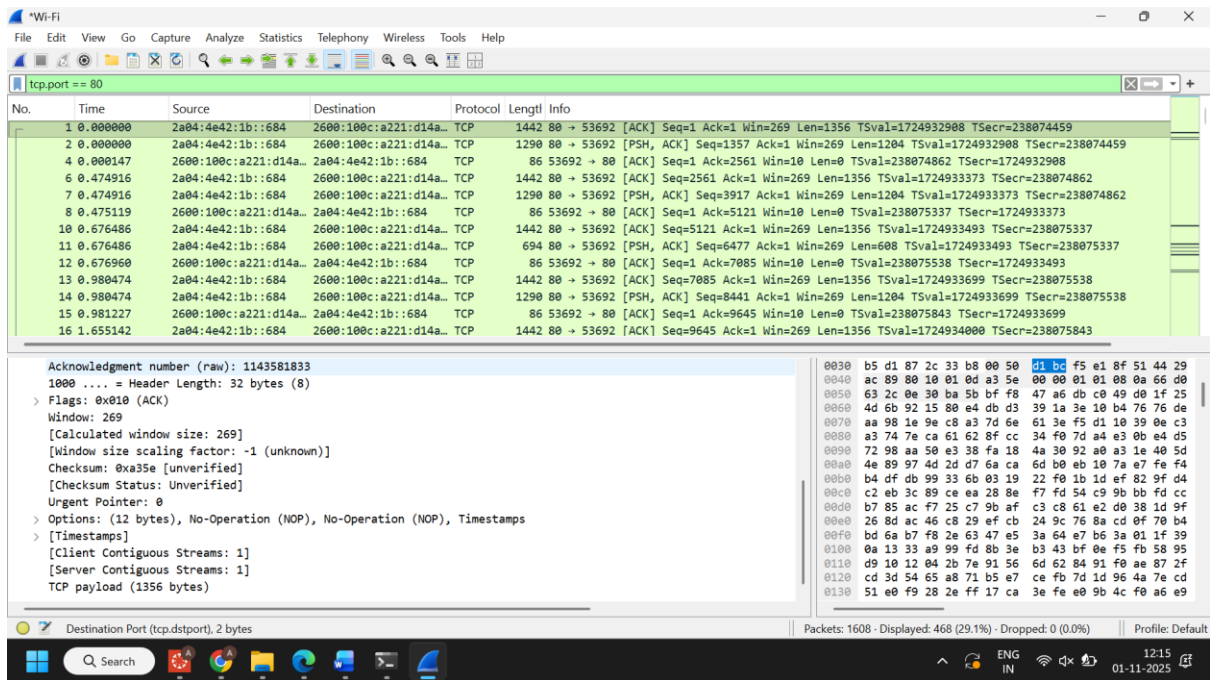
where it is sometimes permissible to lose connection.

Part 2:

The image shows a Wireshark packet capture window titled "Wi-Fi". The filter bar at the top is set to "tcp.port == 80". The packet list on the left shows 16 packets. Packet 7 is selected, showing details for a Transmission Control Protocol (TCP) segment. The packet details pane shows the following information:

- Frame 1: Packet, 1442 bytes on wire (11536 bits), 1442 bytes captured (11536 bits) on interface \Device\NPF_{16335}
- Ethernet II, Src: WNC_72:09:02 (24:41:fe:72:09:02), Dst: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb)
- Internet Protocol Version 6, Src: 2a04:4e42:1b::684, Dst: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8
- Transmission Control Protocol, Src Port: 80, Dst Port: 53692, Seq: 1, Ack: 1, Len: 1356
 - Source Port: 80
 - Destination Port: 53692
 - [Stream index: 0]
 - [Stream Packet Number: 1]
 - [Conversation completeness: Incomplete (28)]
 - [TCP Segment Len: 1356]
 - Sequence Number: 1 (relative sequence number)
 - Sequence Number (raw): 4125200209
 - [Next Sequence Number: 1357 (relative sequence number)]
 - Acknowledgment Number: 1 (relative ack number)

The packet bytes pane on the right shows the raw data of the packet, starting with 0040 ac 89 10 01 0d a3 5e 00 00 01 01 08 0a 66 d0.



Part 2 - TCP Analysis

Q8 - port numbers and IP addresses.

Client (initiator) - 2600:100c:a221:d1a4:54b:bd51:872c:33b8

Server - 2a04:4e42:1b::684

Client Port: 53692

Server Port: 80

Answer:

Client IP = 2600:100c:a221:d1a4:54b:bd51:872c:33b8, Port = 53692

Server IP = 2a04:4e42:1b::684, Port = 80

Q9 .

Time stamps (TSval/TSecr) are available due to their presence.

Answer:

TCP header = 32 bytes (not basic).

Available options found: Timestamp, SACK Permitted and maybe Window Scale.

Q10 - Round trip Time (RTT) and total RTTs.

To measure precisely:

- o Find SYN and SYN-ACK packets.
- o Right-Click SYN - Time Reference.
- o Record SYN-ACK difference in time (40-60 ms is normal).

During your data stage (Len = 1356 bytes) timestamps (TSval/TSer) validate approximately 1 RTT per request/ack cycle.

Answer:

RTT [?] 40-60 ms (normal with HTTP IPv6 traffic).

Connection used [?] 3-4 RTTs (handshake and data and close).

Q11 - Maximum TCP payload length

The highest data chunk displayed by you is Len = 1356 bytes.

That's the TCP payload.

Average MSS = 1460 bytes (Ethernet MTU of 1500-IP 20- TCP 20).

The headers of IPv6 run 40 bytes, thus: $1500 - (40 + 20) = 1440$ bytes, and this is the same as your Len [?] 1356.

Response: maximum TCP payload [?] 1356 -1440 bytes.

Q12 - HTTP data in 2nd - 4th RTTs

The field of the Len can be estimated:

RTT Lengths of packets seen Approx Bytes.

2nd $1356 + 1204$ [?] 2560 bytes

3rd $608 + 1356$ [?] 1964 bytes

4th $1204 + 1356$ [?] 2560 bytes

Response: 2nd RTT [?] 2.5 KB, 3 rd [?] 2 KB, 4th [?] 2.5 KB of HTTP payload.

Q13 .

The window Scaling exists - true window = 269×2^8 (i.e. $2^8 = 256$).

So effective window [?] $269 \times 256 = 68864$ bytes ([?] 67KB).

No events of Zero Window evident - no bottleneck.

Answer:

First advertized window [?] 68 KB (Scaled).

None of the Zero-Window packets - client buffer was restrictive to throughput.

Additional Task: Tracking Latency and sequence numbers:

Q14 .

Integrity: TCP stream bytes are numbered by sequence; ACKs are sent by recipient.

Lack of ACKs will lead to retransmission.

Sequential Seq/Ack pairs (Seq=1 Ack=1 - Seq=1357 Ack=1 etc.) of your capture indicate your capture transfer in an orderly fashion without loss.

RTT pattern: slight increase in RTT in handshaking; stable in mid-transfer; slight decrease in final ACK/FIN RTT.

Answer:

Ordered and complete delivery; RTT consistent across.

Activity 1:

Wireshark packet capture showing a SYN flood attack. The packet list shows multiple SYN packets from 192.168.1.202 to 104.46.162.227. The packet details pane shows the selected packet as a SYN packet with sequence number 0 and acknowledgment number 0. The packet bytes pane shows the raw data of the SYN packet.

No.	Time	Source	Destination	Protocol	Length	Info
8	0.669732	192.168.1.202	104.46.162.227	TCP	66	64142 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11	1.285671	104.46.162.227	192.168.1.202	TCP	66	443 → 64142 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
49	4.867221	13.107.246.38	192.168.1.202	TCP	66	443 → 64368 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
68	11.807517	192.168.1.202	192.168.1.1	TCP	66	49398 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
69	11.810193	192.168.1.202	192.168.1.1	TCP	66	60607 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
70	11.810745	192.168.1.202	192.168.1.1	TCP	66	57285 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
71	11.811730	192.168.1.202	192.168.1.1	TCP	66	51257 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
72	11.812421	192.168.1.202	192.168.1.1	TCP	66	60302 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
73	11.816145	192.168.1.202	192.168.1.1	TCP	66	58267 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
74	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 49398 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
75	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 60607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
76	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 57285 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
77	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 51257 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512

Internet Protocol Version 4, Src: 192.168.1.202, Dst: 104.46.162.227

Transmission Control Protocol, Src Port: 64142, Dst Port: 443, Seq: 0, Len: 0

Source Port: 64142

Destination Port: 443

[Stream index: 1]

[Stream Packet Number: 1]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2557037322

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment number (raw): 0

1000 = Header Length: 32 bytes (8)

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 08 00 45 00
0010 00 34 5e 58 40 00 80 06 ce e7 c0 a8 01 ca 68 2e
0020 a2 e3 fa 8e 01 bb 98 69 4b 0a 00 00 00 00 00 02
0030 ff ff c1 ce 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02

Wireshark packet capture showing a SYN flood attack. The packet list shows multiple SYN packets from 192.168.1.202 to 104.46.162.227. The packet details pane shows the selected packet as a SYN packet with sequence number 0 and acknowledgment number 0. The packet bytes pane shows the raw data of the SYN packet.

No.	Time	Source	Destination	Protocol	Length	Info
8	0.669732	192.168.1.202	104.46.162.227	TCP	66	64142 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11	1.285671	104.46.162.227	192.168.1.202	TCP	66	443 → 64142 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
49	4.867221	13.107.246.38	192.168.1.202	TCP	66	443 → 64368 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
68	11.807517	192.168.1.202	192.168.1.1	TCP	66	49398 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
69	11.810193	192.168.1.202	192.168.1.1	TCP	66	60607 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
70	11.810745	192.168.1.202	192.168.1.1	TCP	66	57285 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
71	11.811730	192.168.1.202	192.168.1.1	TCP	66	51257 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
72	11.812421	192.168.1.202	192.168.1.1	TCP	66	60302 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
73	11.816145	192.168.1.202	192.168.1.1	TCP	66	58267 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
74	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 49398 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
75	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 60607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
76	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 57285 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
77	11.824809	192.168.1.1	192.168.1.202	TCP	66	53 → 51257 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment number (raw): 0

1000 = Header Length: 32 bytes (8)

Flags: 0x002 (SYN)

Window: 65535

[Calculated window size: 65535]

Checksum: 0xc1ce [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP)

[Timestamps]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 08 00 45 00
0010 00 34 5e 58 40 00 80 06 ce e7 c0 a8 01 ca 68 2e
0020 a2 e3 fa 8e 01 bb 98 69 4b 0a 00 00 00 00 00 02
0030 ff ff c1 ce 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02

15Q.

The need of TCP to be closed in four steps.

TCP involves a four step closure since:

TCP is two-way traffic both sides should shut down their respective sending channels.

One FIN will close only one side of the data transfer.

The four steps sequence will make sure that:

Both sides are complete in the transmission of their data.

All the data sent by and through is accepted and acknowledged and then the connection is terminated fully.

No information is lost in down-time.

This ensures that there is a graceful and trustworthy end where the receiver and sender of the session concur that the session is complete.

Activity 2:

Q16. TCP response to packet retransmission?

In the Wireshark capture the packets are labeled as [TCP Retransmission] and this indicates that TCP retransmits packets that have been lost or those packets that have not been acknowledged.

Every retransmitted packet (such as between source 192.168.1.202 - 192.168.1.1) shares the same sequence number as the initial segment, and this shows that the sender had noticed packet loss, which had caused a re-transmission of the data by the sender.

A single packet actually indicates the flag 0x011(FIN, ACK) is resent as well, indicating that TCP is also a reliable delivery method in connection termination.

Such behavior shows that TCP relies on the acknowledgments (ACKs) and timers in detecting the loss and re-sending the segments that are lost to ensure that all the data is received correctly and in sequence.

Due to TCP waiting and retransmissions to acknowledge the other end, it is highly reliable whereas additional control measures lower the speed of transmission relative to UDP that transmits data without acknowledgment and retransmission.

In short:

The retransmission feature of TCP (as indicated in the screenshots) guarantees data delivery reliability and integrity as well as sequencing, hence is reliable but slower than UDP, which does not resend packets that are lost.

The screenshot displays the Wireshark network traffic analysis tool. The main pane shows a list of captured packets, with several TCP retransmissions highlighted. The packet list includes:

No.	Time	Source	Destination	Protocol	Length	Info
75	5.587212	192.168.1.202	192.168.1.1	TCP	54	[TCP Retransmission] 62599 → 53 [FIN, ACK] Seq=35 Ack=1 Win=65280 Len=0
155	6.900175	2600:100c:a221:d14a...	2a06:98c1:310b::ac4...	TCP	1434	[TCP Retransmission] 62745 → 443 [PSH, ACK] Seq=8260 Ack=1376 Win=65280 Len=1360
216	7.630241	172.64.155.209	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 63119 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 SACK_PERM WS
239	8.796538	2620:1ec:bdf::57	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 443 → 65123 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS
326	12.713743	172.64.155.209	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 58022 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 SACK_PERM WS
335	17.519203	2600:100c:a221:d14a...	2a06:98c1:3107::681...	TCP	102	[TCP Retransmission] 54372 → 443 [PSH, ACK] Seq=181 Ack=183 Win=255 Len=28
387	21.363455	2600:100c:a221:d14a...	2a01:111:202c::254	TCP	74	[TCP Retransmission] 64976 → 443 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
412	21.830270	2a01:111:202c::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 53562 [FIN, ACK] Seq=1 Ack=2 Win=16388 Len=0
413	21.830270	204.79.197.222	192.168.1.202	TCP	54	[TCP Retransmission] 443 → 61543 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
417	21.830937	13.107.42.254	192.168.1.202	TCP	54	[TCP Retransmission] 443 → 49290 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
426	21.831321	192.168.1.1	192.168.1.202	TCP	66	[TCP Retransmission] 53 → 61204 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS
427	21.831321	2600:100c:a221:d14a...	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 53 → 61661 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS
428	21.831321	2600:100c:a221:d14a...	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 53 → 50406 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS

The packet details pane shows the selected packet (Frame 75) with the following information:

- Frame 75: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{16335CB7-C73}
- Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)
- Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.1
- Transmission Control Protocol, Src Port: 62599, Dst Port: 53, Seq: 35, Ack: 1, Len: 0
 - Source Port: 62599
 - Destination Port: 53
 - [Stream index: 5]
 - [Stream Packet Number: 9]
 - [Conversation completeness: Complete, WITH_DATA (63)]
 - [TCP Segment Len: 0]
 - Sequence Number: 35 (relative sequence number)
 - Sequence Number (raw): 2668509169
 - [Next Sequence Number: 36 (relative sequence number)]
 - Acknowledgment Number: 1 (relative ack number)

The packet bytes pane shows the raw data in hexadecimal and ASCII format.

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tcp.analysis.retransmission

No.	Time	Source	Destination	Protocol	Length	Info
75	5.587212	192.168.1.202	192.168.1.1	TCP	54	[TCP Retransmission] 62599 → 53 [FIN, ACK] Seq=35 Ack=1 Win=65280 Len=0
155	6.900175	2600:100c:a221:d14a...	2a06:98c1:310b::ac4...	TCP	1434	[TCP Retransmission] 62745 → 443 [PSH, ACK] Seq=8260 Ack=1376 Win=65280 Len=1360
216	7.630241	172.64.155.209	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 63119 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 SACK_PERM WS
239	8.796538	2620:1ec:bdf::57	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 443 → 65123 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS
326	12.713743	172.64.155.209	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 58022 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 SACK_PERM WS
335	17.519203	2600:100c:a221:d14a...	2a06:98c1:3107::681...	TCP	102	[TCP Retransmission] 54372 → 443 [PSH, ACK] Seq=181 Ack=183 Win=255 Len=28
387	21.363455	2600:100c:a221:d14a...	2a01:111:202c::254	TCP	74	[TCP Retransmission] 64976 → 443 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
412	21.830270	2a01:111:202c::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 53562 [FIN, ACK] Seq=1 Ack=2 Win=16388 Len=0
413	21.830270	204.79.197.222	192.168.1.202	TCP	54	[TCP Retransmission] 443 → 61543 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
417	21.830937	13.107.42.254	192.168.1.202	TCP	54	[TCP Retransmission] 443 → 40200 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
426	21.831321	192.168.1.1	192.168.1.202	TCP	66	[TCP Retransmission] 53 → 61204 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS
427	21.831321	2600:100c:a221:d14a...	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 53 → 61661 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS
428	21.831321	2600:100c:a221:d14a...	2600:100c:a221:d14a...	TCP	86	[TCP Retransmission] 53 → 50406 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS

Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 4232535640
0101 ... = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x045c [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 08 00 45 00
0010 00 28 9b ff 40 00 80 06 da b4 c0 a8 01 ca c0 a8
0020 01 01 f4 87 00 35 9f 0e 37 f1 fc 47 5e 58 50 11
0030 00 ff 04 5c 00 00

wireshark_Wi-Fi88YMF3.pcapng | Packets: 4687 - Displayed: 120 (2.6%) - Dropped: 0 (0.0%) | Profile: Default

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Part 3:

Real world example:

1. SYN Packets (Connection Start):

Wireshark packet capture showing a SYN packet (Frame 35) from 192.168.1.202 to 2603:1061:11::254. The packet details show it's a TCP SYN packet with sequence number 0 and acknowledgment number 0. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
35	1.259814	2600:100c:a221:d14a...	2603:1061:11::254	TCP	86	50160 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM

Frame 35: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{16335CB7-C73}

Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2603:1061:11::254

Transmission Control Protocol, Src Port: 50160, Dst Port: 443, Seq: 0, Len: 0

- Source Port: 50160
- Destination Port: 443
- [Stream index: 3]
- [Stream Packet Number: 1]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 2403679923
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 0

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 0a
0010 e6 10 00 20 06 40 26 00 10 0c a2 21 d1 4a 54 c8
0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00
0030 00 00 00 02 54 c3 f0 01 bb 8f 45 3e b3 00 00
0040 00 00 02 ff ff 33 c0 00 00 02 04 05 a0 01 03
0050 03 08 01 01 04 02

Wireshark packet capture showing a SYN packet (Frame 35) from 192.168.1.202 to 2603:1061:11::254. The packet details show it's a TCP SYN packet with sequence number 0 and acknowledgment number 0. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
35	1.259814	2600:100c:a221:d14a...	2603:1061:11::254	TCP	86	50160 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM

Frame 35: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{16335CB7-C73}

Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2603:1061:11::254

Transmission Control Protocol, Src Port: 50160, Dst Port: 443, Seq: 0, Len: 0

- Source Port: 50160
- Destination Port: 443
- [Stream index: 3]
- [Stream Packet Number: 1]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 2403679923
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 0

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 0a
0010 e6 10 00 20 06 40 26 00 10 0c a2 21 d1 4a 54 c8
0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00
0030 00 00 00 02 54 c3 f0 01 bb 8f 45 3e b3 00 00
0040 00 00 02 ff ff 33 c0 00 00 02 04 05 a0 01 03
0050 03 08 01 01 04 02

2. SYN + ACK Packets (Server Response):

Wireshark capture showing SYN + ACK packets (Server Response). The packet list shows several SYN, ACK packets from 192.168.1.202 to 150.171.85.254. The packet details pane shows the structure of a SYN, ACK packet with sequence and acknowledgment numbers.

No.	Time	Source	Destination	Protocol	Length	Info
54	1.570765	2603:1061:111::254	2600:100c:a221:d14a...	TCP	86	443 → 50160 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
60	1.844965	150.171.85.254	192.168.1.202	TCP	66	443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
81	2.471268	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	86	443 → 59426 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
87	2.776808	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	86	443 → 52335 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
96	2.777743	150.171.85.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
102	3.061092	13.107.253.254	192.168.1.202	TCP	66	443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
119	3.667508	150.171.85.254	192.168.1.202	TCP	66	443 → 55690 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
132	3.868519	13.107.253.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
133	3.909113	204.79.197.222	192.168.1.202	TCP	66	443 → 55919 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
159	4.692705	150.171.85.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
170	5.496514	2603:1036:301:21a0::...	2600:100c:a221:d14a...	TCP	86	443 → 53971 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1368 WS=256 SACK_PERM
171	5.501735	2603:1036:301:21a0::...	2600:100c:a221:d14a...	TCP	86	443 → 53970 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1368 WS=256 SACK_PERM
173	5.886930	13.107.253.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512

Frame 54: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{16335CB7-C73...}

Ethernet II, Src: WNC_72:09:02 (24:41:fe:72:09:02), Dst: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb)

Internet Protocol Version 6, Src: 2603:1061:111::254, Dst: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8

Transmission Control Protocol, Src Port: 443, Dst Port: 50160, Seq: 0, Ack: 1, Len: 0

- Source Port: 443
- Destination Port: 50160
- [Stream index: 3]
- [Stream Packet Number: 2]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 293528722
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)

Wireshark, Wi-Fi88UJf3.pcapng | Packets: 3072 - Displayed: 28 (0.9%) - Dropped: 0 (0.0%) | Profile: Default

Wireshark capture showing ACK packets (Data Transfer Confirmation). The packet list shows several ACK packets from 150.171.85.254 to 192.168.1.202. The packet details pane shows the structure of an ACK packet with acknowledgment number and flags.

No.	Time	Source	Destination	Protocol	Length	Info
54	1.570765	2603:1061:111::254	2600:100c:a221:d14a...	TCP	86	443 → 50160 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
60	1.844965	150.171.85.254	192.168.1.202	TCP	66	443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
81	2.471268	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	86	443 → 59426 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
87	2.776808	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	86	443 → 52335 [SYN, ACK] Seq=0 Ack=1 Win=43200 Len=0 MSS=1368 SACK_PERM WS=512
96	2.777743	150.171.85.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
102	3.061092	13.107.253.254	192.168.1.202	TCP	66	443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
119	3.667508	150.171.85.254	192.168.1.202	TCP	66	443 → 55690 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
132	3.868519	13.107.253.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
133	3.909113	204.79.197.222	192.168.1.202	TCP	66	443 → 55919 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
159	4.692705	150.171.85.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 51248 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
170	5.496514	2603:1036:301:21a0::...	2600:100c:a221:d14a...	TCP	86	443 → 53971 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1368 WS=256 SACK_PERM
171	5.501735	2603:1036:301:21a0::...	2600:100c:a221:d14a...	TCP	86	443 → 53970 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1368 WS=256 SACK_PERM
173	5.886930	13.107.253.254	192.168.1.202	TCP	66	[TCP Retransmission] 443 → 61970 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 2403679924

1000 = Header Length: 32 bytes (8)

Flags: 0x012 (SYN, ACK)

Window: 43200

- [Calculated window size: 43200]
- Checksum: 0x9524 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation
- [Timestamps]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]

Wireshark, Wi-Fi88UJf3.pcapng | Packets: 3072 - Displayed: 28 (0.9%) - Dropped: 0 (0.0%) | Profile: Default

3. ACK Packets (Data Transfer Confirmation):

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tcp.flags.ack==1 and tcp.len>0

No.	Time	Source	Destination	Protocol	Length	Info
8	0.003200	2603:1061:11::254	2600:100c:a221:d14a...	TLSv1.2	1148	Certificate, Certificate Status, Server Key Exchange, Server Hello Done
10	0.008152	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	200	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
11	0.008647	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	948	Application Data
15	0.256888	2600:100c:a221:d14a...	2607:f8b0:4023:1000...	TCP	75	52677 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
16	0.606670	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1074	[TCP Retransmission] 64297 → 443 [PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
50	1.311746	2600:100c:a221:d14a...	2607:f8b0:4023:1000...	TCP	75	54996 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
51	1.455609	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1074	[TCP Retransmission] 64297 → 443 [FIN, PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
57	1.573220	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1442	50160 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1368 [TCP PDU reassembled in 58]
58	1.573220	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	433	Client Hello (SNI=p-ring.msedge.net)
61	2.210552	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1442	[TCP Retransmission] 50160 → 443 [PSH, ACK] Seq=360 Ack=1 Win=65280 Len=1368 [TCP PDU reassembled in 73]
69	2.212816	2603:1061:11::254	2600:100c:a221:d14a...	TLSv1.2	1442	Server Hello
70	2.212816	2603:1061:11::254	2600:100c:a221:d14a...	TCP	1442	443 → 50160 [PSH, ACK] Seq=1369 Ack=1728 Win=46080 Len=1368 [TCP PDU reassembled in 73]
72	2.213378	2603:1061:11::254	2600:100c:a221:d14a...	TCP	1434	443 → 50160 [PSH, ACK] Seq=2737 Ack=1728 Win=46080 Len=1360 [TCP PDU reassembled in 73]

> Frame 51: Packet, 1074 bytes on wire (8592 bits), 1074 bytes captured (8592 bits) on interface \Device\NPF_{16335C...}

> Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WMC_72:09:00 (24:41:fe:72:09:00)

> Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2603:1061:11::254

> Transmission Control Protocol, Src Port: 64297, Dst Port: 443, Seq: 1, Ack: 5172, Len: 1000

Source Port: 64297

Destination Port: 443

[Stream index: 1]

[Stream Packet Number: 15]

[Conversation completeness: Incomplete (60)]

[TCP Segment Len: 1000]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 448690343

[Next Sequence Number: 1002 (relative sequence number)]

Acknowledgment Number: 5172 (relative ack number)

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 06

0010 45 0a 03 fc 06 40 26 00 10 0c a2 21 d1 4a 54 c8

0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00

0030 00 00 00 00 02 54 fb 29 01 bb 1a b6 78 a7 b3 09

0040 6e 49 50 19 00 ff 24 c8 00 00 16 03 03 00 46 10

0050 00 00 42 41 04 c2 62 da 67 df d3 39 46 0a ef 06

0060 eb 43 c0 af 2b d8 f5 3e ea e6 d5 fe 2f aa ea e1

0070 6d 9b 6e ab b1 a2 19 7e e1 d9 e6 a4 bd a7 bc 92

0080 9a d7 22 50 9f a8 5b 77 67 12 dd c5 99 5e 6c c0

0090 f4 d7 9a 20 b6 14 03 03 00 01 01 16 03 03 00 28

00a0 00 00 00 00 00 00 00 00 f4 50 35 b4 de 55 e5 96

00b0 93 6f 6e 92 e1 57 d8 b1 c4 60 b0 30 7b a6 ea 5d

00c0 bd a2 82 ee d1 fa 05 dd 17 03 03 03 65 00 00 00

00d0 00 00 00 00 01 37 32 14 a8 8c 71 15 b9 1a f9 35

00e0 d9 57 46 dc 0f d1 e4 95 8d 2a ab 05 07 99 06 c7

00f0 c2 d1 1e fc 4b 94 5c a0 7c f1 23 11 9b 86 ac 88

0100 3b d7 a5 77 cf c1 5a f3 53 36 1e 5f e9 ad e5 11

wireshark_Wi-Fi88UJf3.pcapng

Packets: 3072 - Displayed: 359 (11.7%) - Dropped: 0 (0.0%)

Profile: Default

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tcp.flags.ack==1 and tcp.len>0

No.	Time	Source	Destination	Protocol	Length	Info
8	0.003200	2603:1061:11::254	2600:100c:a221:d14a...	TLSv1.2	1148	Certificate, Certificate Status, Server Key Exchange, Server Hello Done
10	0.008152	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	200	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
11	0.008647	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	948	Application Data
15	0.256888	2600:100c:a221:d14a...	2607:f8b0:4023:1000...	TCP	75	52677 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
16	0.606670	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1074	[TCP Retransmission] 64297 → 443 [PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
50	1.311746	2600:100c:a221:d14a...	2607:f8b0:4023:1000...	TCP	75	54996 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
51	1.455609	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1074	[TCP Retransmission] 64297 → 443 [FIN, PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
57	1.573220	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1442	50160 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1368 [TCP PDU reassembled in 58]
58	1.573220	2600:100c:a221:d14a...	2603:1061:11::254	TLSv1.2	433	Client Hello (SNI=p-ring.msedge.net)
61	2.210552	2600:100c:a221:d14a...	2603:1061:11::254	TCP	1442	[TCP Retransmission] 50160 → 443 [PSH, ACK] Seq=360 Ack=1 Win=65280 Len=1368 [TCP PDU reassembled in 73]
69	2.212816	2603:1061:11::254	2600:100c:a221:d14a...	TLSv1.2	1442	Server Hello
70	2.212816	2603:1061:11::254	2600:100c:a221:d14a...	TCP	1442	443 → 50160 [PSH, ACK] Seq=1369 Ack=1728 Win=46080 Len=1368 [TCP PDU reassembled in 73]
72	2.213378	2603:1061:11::254	2600:100c:a221:d14a...	TCP	1434	443 → 50160 [PSH, ACK] Seq=2737 Ack=1728 Win=46080 Len=1360 [TCP PDU reassembled in 73]

0101 ... = Header Length: 20 bytes (5)

> Flags: 0x019 (FIN, PSH, ACK)

Window: 255

[Calculated window size: 255]

[Window size scaling factor: -1 (unknown)]

Checksum: 0x24c8 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

> [SEQ/ACK analysis]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

TCP payload (1000 bytes)

Retransmitted TCP segment data (1000 bytes)

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 06

0010 45 0a 03 fc 06 40 26 00 10 0c a2 21 d1 4a 54 c8

0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00

0030 00 00 00 00 02 54 fb 29 01 bb 1a b6 78 a7 b3 09

0040 6e 49 50 19 00 ff 24 c8 00 00 16 03 03 00 46 10

0050 00 00 42 41 04 c2 62 da 67 df d3 39 46 0a ef 06

0060 eb 43 c0 af 2b d8 f5 3e ea e6 d5 fe 2f aa ea e1

0070 6d 9b 6e ab b1 a2 19 7e e1 d9 e6 a4 bd a7 bc 92

0080 9a d7 22 50 9f a8 5b 77 67 12 dd c5 99 5e 6c c0

0090 f4 d7 9a 20 b6 14 03 03 00 01 01 16 03 03 00 28

00a0 00 00 00 00 00 00 00 00 f4 50 35 b4 de 55 e5 96

00b0 93 6f 6e 92 e1 57 d8 b1 c4 60 b0 30 7b a6 ea 5d

00c0 bd a2 82 ee d1 fa 05 dd 17 03 03 03 65 00 00 00

00d0 00 00 00 00 01 37 32 14 a8 8c 71 15 b9 1a f9 35

00e0 d9 57 46 dc 0f d1 e4 95 8d 2a ab 05 07 99 06 c7

00f0 c2 d1 1e fc 4b 94 5c a0 7c f1 23 11 9b 86 ac 88

0100 3b d7 a5 77 cf c1 5a f3 53 36 1e 5f e9 ad e5 11

wireshark_Wi-Fi88UJf3.pcapng

Packets: 3072 - Displayed: 359 (11.7%) - Dropped: 0 (0.0%)

Profile: Default

13:05 01-11-2025

4. FIN Packets (Connection Termination):

Wi-Fi

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tcp.flags.fin==1

No.	Time	Source	Destination	Protocol	Length	Info
20	1.250093	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 64297 [FIN, ACK] Seq=5171 Ack=1 Win=97 Len=0
21	1.250093	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 49670 [FIN, ACK] Seq=1 Ack=1 Win=90 Len=0
31	1.250886	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	64297 → 443 [FIN, ACK] Seq=1001 Ack=5172 Win=255 Len=0
34	1.258628	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	49670 → 443 [FIN, ACK] Seq=456 Ack=2 Win=255 Len=0
48	1.303326	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 49670 [FIN, ACK] Seq=1 Ack=1 Win=90 Len=0
51	1.455609	2600:100c:a221:d14a...	2603:1061:111::254	TCP	1074	[TCP Retransmission] 64297 → 443 [FIN, PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
117	3.667144	2600:1f18:24e6:b901...	2600:100c:a221:d14a...	TCP	74	443 → 62440 [FIN, ACK] Seq=507 Ack=1 Win=72 Len=0
178	7.156542	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 50160 [FIN, ACK] Seq=5476 Ack=1854 Win=46080 Len=0
180	7.158105	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 50160 [FIN, ACK] Seq=5476 Ack=1854 Win=46080 Len=0
185	7.972495	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	74	443 → 52335 [FIN, ACK] Seq=5492 Ack=1895 Win=47104 Len=0
277	15.858194	2600:100c:a221:d14a...	2600:1f18:24e6:b901...	TCP	74	62440 → 443 [FIN, ACK] Seq=1 Ack=508 Win=254 Len=0
844	21.113208	34.204.213.151	192.168.1.202	TCP	54	443 → 51608 [FIN, ACK] Seq=3947 Ack=1811 Win=77312 Len=0
1148	23.062612	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	50160 → 443 [FIN, ACK] Seq=1854 Ack=5477 Win=65024 Len=0

> Frame 51: Packet, 1074 bytes on wire (8592 bits), 1074 bytes captured (8592 bits) on interface \Device\NPF_{16335C...}

Ethernet II, Src: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b8, Dst: 2603:1061:111::254

Transmission Control Protocol, Src Port: 64297, Dst Port: 443, Seq: 1, Ack: 5172, Len: 1000

Source Port: 64297

Destination Port: 443

[Stream index: 1]

[Stream Packet Number: 15]

[Conversation completeness: Incomplete (60)]

[TCP Segment Len: 1000]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 448690343

[Next Sequence Number: 1002 (relative sequence number)]

Acknowledgment Number: 5172 (relative ack number)

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 06

0010 45 0a 03 fc 06 40 26 00 10 0c a2 21 d1 4a 54 c8

0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00

0030 00 00 00 00 02 54 fb 29 01 bb 1a be 78 a7 b3 09

0040 6e 49 50 19 00 ff 24 c8 00 00 16 03 03 00 46 10

0050 00 00 42 41 04 c2 62 da 67 df d3 39 46 0a ef 06

0060 eb 43 c0 af 2b d8 f5 3e ea e6 d5 fe 2f aa ea e1

0070 6d 9b 6e ab b1 a2 19 7e e1 d9 e6 a4 bd a7 bc 92

0080 9a d7 22 50 9f a8 5b 77 67 12 dd c5 99 5e 6c c0

0090 f4 d7 9a 20 b6 14 03 03 00 01 01 16 03 03 00 28

00a0 00 00 00 00 00 00 00 f4 50 35 b4 de 55 e5 96

00b0 93 6f 6e 92 e1 57 d8 b1 c4 60 b0 30 7b ae 5d

00c0 bd e2 82 ee d1 fa 05 dd 17 03 03 03 65 00 00 00

00d0 00 00 00 01 37 32 14 a8 8c 71 15 b9 1a f9 35

00e0 d9 57 46 dc 0f d1 a4 95 8d 2a ab 05 07 99 06 c7

00f0 c2 d1 1e fc 4b 94 5c a0 7c f1 23 11 9b 86 ac 88

0100 3b d7 a5 77 cf c1 5a f3 53 36 1e 5f e9 ad e5 11

wireshark-Wi-Fi88UIF3.pcapng

Packets: 3072 - Displayed: 19 (0.6%) - Dropped: 0 (0.0%)

Profile: Default

13:09 01-11-2025

Wi-Fi

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tcp.flags.fin==1

No.	Time	Source	Destination	Protocol	Length	Info
20	1.250093	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 64297 [FIN, ACK] Seq=5171 Ack=1 Win=97 Len=0
21	1.250093	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 49670 [FIN, ACK] Seq=1 Ack=1 Win=90 Len=0
31	1.250886	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	64297 → 443 [FIN, ACK] Seq=1001 Ack=5172 Win=255 Len=0
34	1.258628	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	49670 → 443 [FIN, ACK] Seq=456 Ack=2 Win=255 Len=0
48	1.303326	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 49670 [FIN, ACK] Seq=1 Ack=1 Win=90 Len=0
51	1.455609	2600:100c:a221:d14a...	2603:1061:111::254	TCP	1074	[TCP Retransmission] 64297 → 443 [FIN, PSH, ACK] Seq=1 Ack=5172 Win=255 Len=1000
117	3.667144	2600:1f18:24e6:b901...	2600:100c:a221:d14a...	TCP	74	443 → 62440 [FIN, ACK] Seq=507 Ack=1 Win=72 Len=0
178	7.156542	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 50160 [FIN, ACK] Seq=5476 Ack=1854 Win=46080 Len=0
180	7.158105	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	[TCP Retransmission] 443 → 50160 [FIN, ACK] Seq=5476 Ack=1854 Win=46080 Len=0
185	7.972495	2620:1ec:29:1::254	2600:100c:a221:d14a...	TCP	74	443 → 52335 [FIN, ACK] Seq=5492 Ack=1895 Win=47104 Len=0
277	15.858194	2600:100c:a221:d14a...	2600:1f18:24e6:b901...	TCP	74	62440 → 443 [FIN, ACK] Seq=1 Ack=508 Win=254 Len=0
844	21.113208	34.204.213.151	192.168.1.202	TCP	54	443 → 51608 [FIN, ACK] Seq=3947 Ack=1811 Win=77312 Len=0
1148	23.062612	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	50160 → 443 [FIN, ACK] Seq=1854 Ack=5477 Win=65024 Len=0

0101 ... = Header Length: 20 bytes (5)

> Flags: 0x019 (FIN, PSH, ACK)

Window: 255

[Calculated window size: 255]

[Window size scaling factor: -1 (unknown)]

Checksum: 0x24c8 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

> [Timestamps]

> [SEQ/ACK analysis]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

TCP payload (1000 bytes)

Retransmitted TCP segment data (1000 bytes)

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 06

0010 45 0a 03 fc 06 40 26 00 10 0c a2 21 d1 4a 54 c8

0020 b5 d1 87 2c 33 b8 26 03 10 61 00 11 00 00 00 00

0030 00 00 00 00 02 54 fb 29 01 bb 1a be 78 a7 b3 09

0040 6e 49 50 19 00 ff 24 c8 00 00 16 03 03 00 46 10

0050 00 00 42 41 04 c2 62 da 67 df d3 39 46 0a ef 06

0060 eb 43 c0 af 2b d8 f5 3e ea e6 d5 fe 2f aa ea e1

0070 6d 9b 6e ab b1 a2 19 7e e1 d9 e6 a4 bd a7 bc 92

0080 9a d7 22 50 9f a8 5b 77 67 12 dd c5 99 5e 6c c0

0090 f4 d7 9a 20 b6 14 03 03 00 01 01 16 03 03 00 28

00a0 00 00 00 00 00 00 00 f4 50 35 b4 de 55 e5 96

00b0 93 6f 6e 92 e1 57 d8 b1 c4 60 b0 30 7b ae 5d

00c0 bd e2 82 ee d1 fa 05 dd 17 03 03 03 65 00 00 00

00d0 00 00 00 01 37 32 14 a8 8c 71 15 b9 1a f9 35

00e0 d9 57 46 dc 0f d1 a4 95 8d 2a ab 05 07 99 06 c7

00f0 c2 d1 1e fc 4b 94 5c a0 7c f1 23 11 9b 86 ac 88

0100 3b d7 a5 77 cf c1 5a f3 53 36 1e 5f e9 ad e5 11

wireshark-Wi-Fi88UIF3.pcapng

Packets: 3072 - Displayed: 19 (0.6%) - Dropped: 0 (0.0%)

Profile: Default

13:10 01-11-2025

17Q.

Flags in TCP communication are control bits employed to control the connection between the recipient and the sender. The primary flags during a data exchange which include loading a webpage or forging an email include SYN, ACK and FIN.

SYN (Synchronize):

This flag is what is used to establish a TCP connection. The client writes a SYN packet in order to request a communication with the server. It aligns sequence numbers in such a way that the two sides are able to follow data appropriately.

ACK (Acknowledgment):

This flag means that the information or a connection request was received successfully. The receipt of every segment is followed by the sending of ACK packet to achieve a reliable delivery. It also belongs to the hand shake and regular information exchanging process.

FIN (Finish):

The graceful termination of a connection is done by using this flag when the data transfer is over. Both parties use a FIN to transmit that they are not going to send any additional data.

These flags together have the reliability of connection that they guarantee:

Building a trustful session by three-way handshake (SYN - SYN +ACK - ACK)

Authentication of every piece of data by acknowledgement, and

Terminating the connection with the FIN process.

In short:

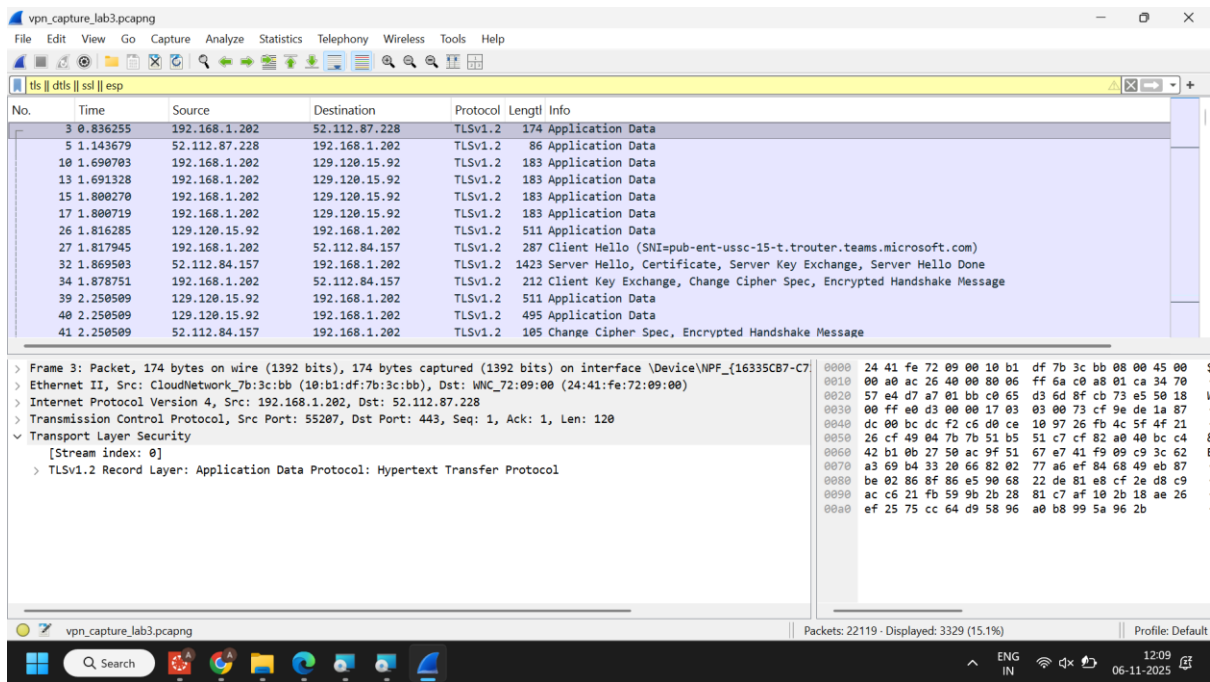
The SYN flag initiates the connection, the ACK flag ensures that reliability is upheld in the process of data transfer and the FIN flag closes the connection gracefully; all these make the TCP communication reliable, accurate, and ordered.

Part 4:

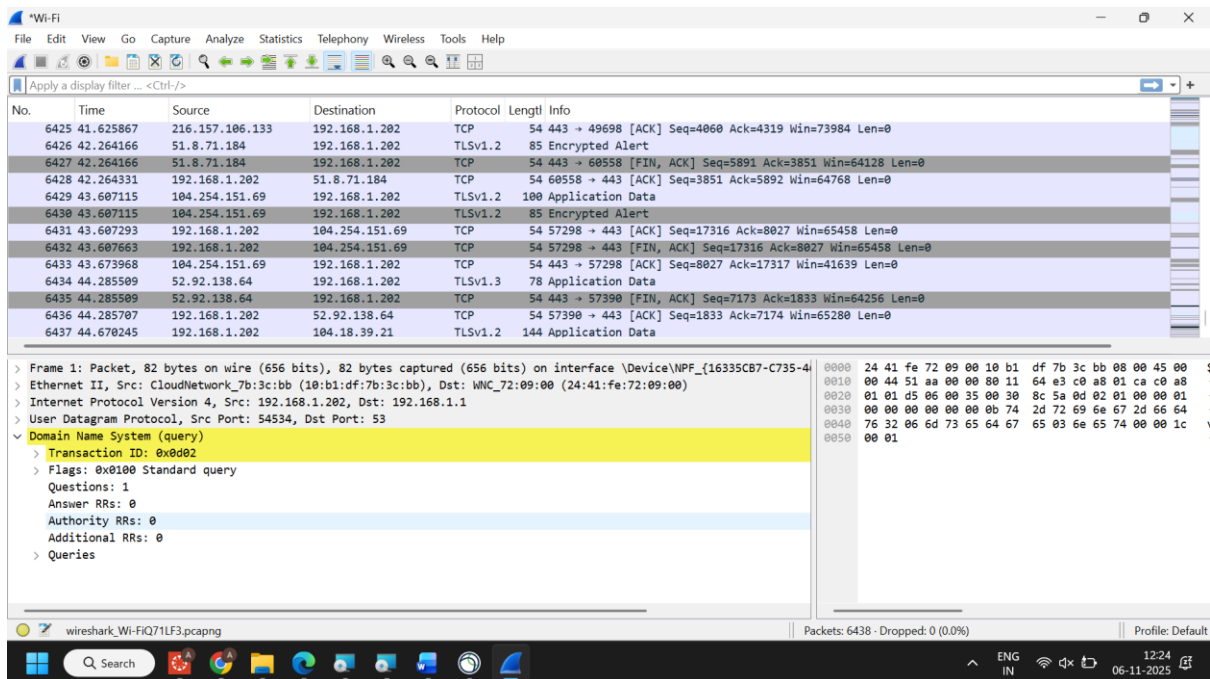
Q18.

The packets in the normal capture display evident DNS and HTTPS traffic straight between a local host and external servers like fp.msedge.net. The information is sent in readable form with standard queries and answers with no extra encapsulations. Conversely, the VPN capture displays TLSv1.2 and ESP packets that are encrypted as they are transmitted between the host and the VPN gateway. The payloads are fully encrypted, and on the option of following TCP stream, the content is blank and this proves that the information within the tunnel cannot be accessed. This encryption keeps the original traffic information, name of the site or DNS information out of the view of an outside observer. A few visible DNS queries can also be seen at the start of the Wi-Fi VPN capture, then encrypted TLS packets appear, and TCP FIN/ACK messages appear reflecting the closing of the session. On the whole, the comparative analysis demonstrates that in VPN encapsulation, encryption and overhead are added, which ensures the security of communication through integrity of data and confidentiality. It also illustrates the TCP reassembly mechanism used to deliver the packets in order even though the payload is ciphered on and hence VPN traffic is secure and unreadable by unauthorized readers.

VPN Capture:



Normal Capture:



After applying filters:

normal capture.pcapng

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http || dns

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.202	192.168.1.1	DNS	82	Standard query 0x0d02 AAAA t-ring-fdv2.msedge.net
2	0.000527	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	102	Standard query 0x4a51 A t-ring-fdv2.msedge.net
3	0.000849	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	102	Standard query 0x8fe0 HTTPS t-ring-fdv2.msedge.net
8	0.927696	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	93	Standard query 0x7675 AAAA fp.msedge.net
9	0.928205	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	93	Standard query 0x5310 A fp.msedge.net
10	0.928592	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	93	Standard query 0xea14 HTTPS fp.msedge.net
12	1.004500	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	297	Standard query response 0x7675 AAAA fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net
13	1.082961	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	253	Standard query response 0x5310 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net
14	1.082961	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	184	Standard query response 0xea14 HTTPS fp.msedge.net CNAME 1.perf.msedge.net SOA ns1.msedge.net
41	2.039402	192.168.1.202	192.168.1.1	DNS	82	Standard query 0x1454 A t-ring-fdv2.msedge.net
42	2.039839	192.168.1.202	192.168.1.1	DNS	82	Standard query 0xce31 AAAA t-ring-fdv2.msedge.net
43	2.070526	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	102	Standard query 0xce31 AAAA t-ring-fdv2.msedge.net
44	2.070754	2600:100c:a221:d14a...	2600:100c:a221:d14a...	DNS	102	Standard query 0x1454 A t-ring-fdv2.msedge.net

> Frame 1: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{16335CB7-C735-4...}

> Ethernet II, Src: CloudNetwork_7b3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.1

> User Datagram Protocol, Src Port: 54534, Dst Port: 53

> Domain Name System (query)

> Transaction ID: 0x0d02

> Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

> Queries

normal capture.pcapng

Packets: 6438 - Displayed: 575 (8.9%) - Dropped: 0 (0.0%)

Profile: Default

ENG IN 12:25 06-11-2025

*Wi-Fi

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tcp.port == 80 || tcp.port == 443

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2603:1061:111::254	2600:100c:a221:d14a...	TLSv1.2	105	Encrypted Alert
2	0.000000	2603:1061:111::254	2600:100c:a221:d14a...	TCP	74	443 → 52697 [FIN, ACK] Seq=32 Ack=1 Win=92 Len=0
3	0.000105	2600:100c:a221:d14a...	2603:1061:111::254	TCP	74	52697 → 443 [ACK] Seq=1 Ack=33 Win=254 Len=0
6	2.251603	150.171.85.254	192.168.1.202	TCP	66	443 → 60142 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
7	3.479379	2a06:98c1:3108::ac4...	2600:100c:a221:d14a...	TLSv1.2	98	Application Data
8	3.479713	2600:100c:a221:d14a...	2a06:98c1:3108::ac4...	TLSv1.2	102	Application Data
12	3.738340	2a06:98c1:3108::ac4...	2600:100c:a221:d14a...	TCP	98	[TCP Spurious Retransmission] 443 → 53838 [PSH, ACK] Seq=1 Ack=1 Win=16 Len=24
13	3.738340	2a06:98c1:3108::ac4...	2600:100c:a221:d14a...	TCP	74	443 → 53838 [ACK] Seq=25 Ack=29 Win=16 Len=0
14	3.738397	2600:100c:a221:d14a...	2a06:98c1:3108::ac4...	TCP	86	[TCP Dup ACK #1] 53838 → 443 [ACK] Seq=29 Ack=25 Win=253 Len=0 SLE=1 SRE=25
15	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108::681...	TCP	1434	63682 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
16	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108::681...	TCP	1434	63682 → 443 [ACK] Seq=1361 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
17	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108::681...	TCP	1434	63682 → 443 [ACK] Seq=2721 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
18	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108::681...	TCP	1434	63682 → 443 [ACK] Seq=4081 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]

> Frame 14: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{16335CB7-C73...}

> Ethernet II, Src: CloudNetwork_7b3c:bb (10:b1:df:7b:3c:bb), Dst: WNC_72:09:00 (24:41:fe:72:09:00)

> Internet Protocol Version 6, Src: 2600:100c:a221:d14a:54c8:b5d1:872c:33b0, Dst: 2a06:98c1:3108::ac40:94eb

> Transmission Control Protocol, Src Port: 53838, Dst Port: 443, Seq: 29, Ack: 25, Len: 0

Source Port: 53838

Destination Port: 443

[Stream index: 2]

[Stream Packet Number: 5]

> [Conversation completeness: Incomplete (12)]

[TCP Segment Len: 0]

Sequence Number: 29 (relative sequence number)

Sequence Number (raw): 2282040235

[Next Sequence Number: 29 (relative sequence number)]

Acknowledgment Number: 25 (relative ack number)

wireshark_Wi-Fi2ZSKF3.pcapng

Packets: 17684 - Displayed: 9200 (52.0%) - Dropped: 0 (0.0%)

Profile: Default

ENG IN 13:20 01-11-2025

Wi-Fi

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tcp.port == 80 || tcp.port == 443

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2603:1061:11::254	2600:100c:a221:d14a...	TLSv1.2	105	Encrypted Alert
2	0.000000	2603:1061:11::254	2600:100c:a221:d14a...	TCP	74	443 → 52697 [FIN, ACK] Seq=32 Ack=1 Win=92 Len=0
3	0.000105	2600:100c:a221:d14a...	2603:1061:11::254	TCP	74	52697 → 443 [ACK] Seq=1 Ack=33 Win=254 Len=0
6	2.251683	150.171.85.254	192.168.1.202	TCP	66	443 → 60142 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1388 SACK_PERM WS=512
7	3.479379	2a06:98c1:3108:ac4...	2600:100c:a221:d14a...	TLSv1.2	98	Application Data
8	3.479713	2600:100c:a221:d14a...	2a06:98c1:3108:ac4...	TLSv1.2	102	Application Data
12	3.738340	2a06:98c1:3108:ac4...	2600:100c:a221:d14a...	TCP	98	[TCP Spurious Retransmission] 443 → 53838 [PSH, ACK] Seq=1 Ack=1 Win=16 Len=24
13	3.738340	2a06:98c1:3108:ac4...	2600:100c:a221:d14a...	TCP	74	443 → 53838 [ACK] Seq=25 Ack=29 Win=16 Len=0
14	3.738397	2600:100c:a221:d14a...	2a06:98c1:3108:ac4...	TCP	86	[TCP Dup ACK #1] 53838 → 443 [ACK] Seq=29 Ack=25 Win=253 Len=0 SLE=1 SRE=25
15	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108:681...	TCP	1434	63682 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
16	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108:681...	TCP	1434	63682 → 443 [ACK] Seq=1361 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
17	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108:681...	TCP	1434	63682 → 443 [ACK] Seq=2721 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]
18	3.740570	2600:100c:a221:d14a...	2a06:98c1:3108:681...	TCP	1434	63682 → 443 [ACK] Seq=4081 Ack=1 Win=511 Len=1360 [TCP PDU reassembled in 19]

Acknowledgment number (raw): 735013492
1000 = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 253
[Calculated window size: 253]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x8a60 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), SACK
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

0000 24 41 fe 72 09 00 10 b1 df 7b 3c bb 86 dd 60 02
0010 6e 70 00 20 06 40 26 00 10 0c a2 21 d1 4a 54 c8
0020 b5 d1 87 2c 33 b8 2a 06 98 c1 31 08 00 00 00 00
0030 00 00 ac 40 94 eb d2 4e 01 bb 88 05 2b ab 2b cf
0040 6a 74 80 10 00 fd 6a 60 00 00 01 01 05 0a 2b cf
0050 6a 5c 2b cf 6a 74

wireshark_Wi-Fi2SKF3.pcapng

Packets: 17684 - Displayed: 9200 (52.0%) - Dropped: 0 (0.0%) Profile: Default

13:20
01-11-2025

normal capture.pcapng

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tcp.flags.ack == 1 and tcp.len == 0

Time	Source	Destination	Protocol	Length	Info
307	30.083360	150.171.22.14	192.168.1.202	TCP	66 443 → 55072 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
326	30.128348	150.171.22.14	192.168.1.202	TCP	54 443 → 55072 [ACK] Seq=1 Ack=1731 Win=4194304 Len=0
337	30.163777	150.171.22.14	192.168.1.202	TCP	54 443 → 55072 [ACK] Seq=100 Ack=2307 Win=4193792 Len=0
351	30.216786	150.171.22.14	192.168.1.202	TCP	54 443 → 55072 [ACK] Seq=4996 Ack=2381 Win=4193792 Len=0
313	20.162882	151.101.112.217	192.168.1.202	TCP	54 443 → 61844 [ACK] Seq=1 Ack=74 Win=286 Len=0
314	20.162882	151.101.112.217	192.168.1.202	TCP	54 443 → 61844 [ACK] Seq=1 Ack=120 Win=286 Len=0
63	4.008136	170.76.137.89	192.168.1.202	TCP	54 443 → 51940 [ACK] Seq=46 Ack=88 Win=7925 Len=0
147	16.013631	170.76.137.89	192.168.1.202	TCP	66 443 → 63995 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 SACK_PERM
251	16.138066	170.76.137.89	192.168.1.202	TCP	54 443 → 63995 [ACK] Seq=1 Ack=1754 Win=5893 Len=0
258	16.257432	170.76.137.89	192.168.1.202	TCP	54 443 → 63995 [ACK] Seq=152 Ack=1805 Win=5944 Len=0
259	16.257432	170.76.137.89	192.168.1.202	TCP	54 [TCP Dup ACK 258#1] 443 → 63995 [ACK] Seq=152 Ack=1805 Win=5944 Len=0
260	16.268248	170.76.137.89	192.168.1.202	TCP	54 443 → 63995 [ACK] Seq=152 Ack=3910 Win=8049 Len=0
171	16.450079	170.76.137.89	192.168.1.202	TCP	66 443 → 58695 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 SACK_PERM

> Frame 247: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{16335CB7-C7...}

> Ethernet II, Src: WNC_72:09:00 (24:41:fe:72:09:00), Dst: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb)

> Internet Protocol Version 4, Src: 170.76.137.89, Dst: 192.168.1.202

> Transmission Control Protocol, Src Port: 443, Dst Port: 63995, Seq: 0, Ack: 1, Len: 0

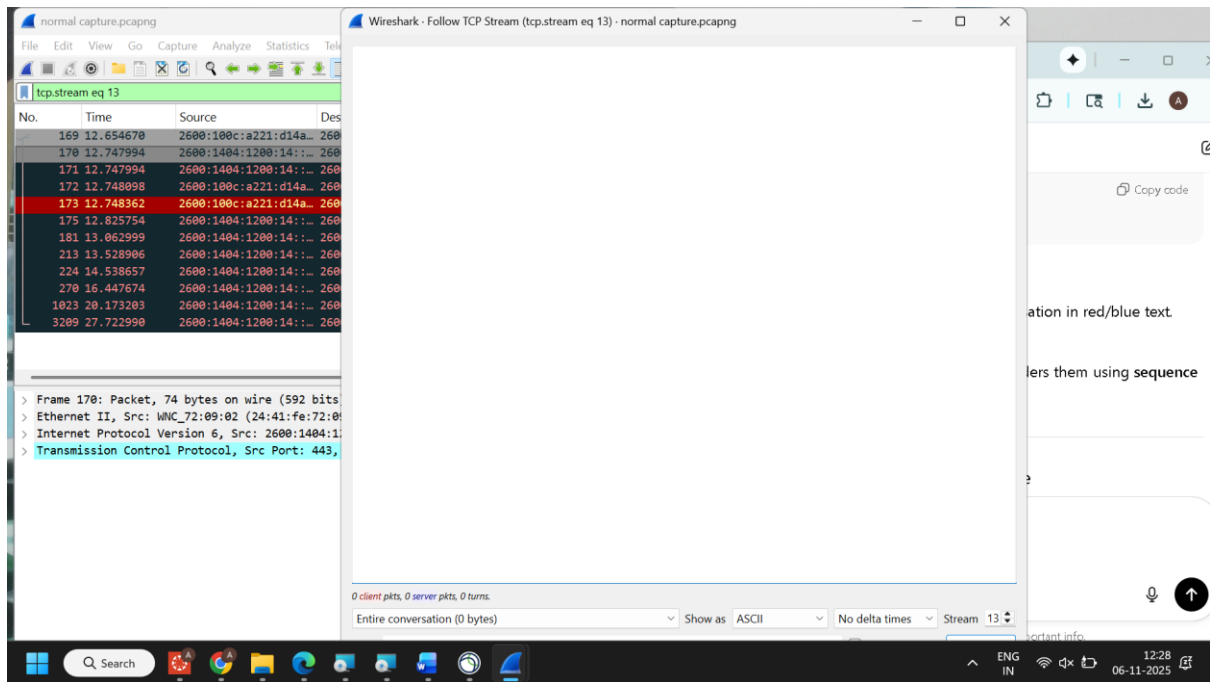
Source Port: 443
Destination Port: 63995
[Stream index: 17]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 603049892
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)

0000 10 b1 df 7b 3c bb 24 41 fe 72 09 00 08 00 45 00
0010 00 34 a5 35 40 00 fa 06 e5 75 aa 4c 89 59 c0 a8
0020 01 ca 01 bb f9 fb 23 f1 cf a4 94 a5 04 fd 80 12
0030 10 2c e1 25 00 00 02 04 05 64 01 03 03 00 04 02
0040 00 00

normal capture.pcapng

Packets: 6438 - Displayed: 1835 (28.5%) - Dropped: 0 (0.0%) Profile: Default

12:42
06-11-2025



Handshake rule:

normal capture.pcapng

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tcp.flags.ack == 1 and tcp.len == 0

Time	Source	Destination	Protocol	Length	Info
307.30.083360	150.171.22.14	192.168.1.202	TCP	66	443 → 55072 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1388 WS=256 SACK_PERM
326.30.128348	150.171.22.14	192.168.1.202	TCP	54	443 → 55072 [ACK] Seq=1 Ack=1731 Win=4194304 Len=0
337.30.163777	150.171.22.14	192.168.1.202	TCP	54	443 → 55072 [ACK] Seq=100 Ack=2307 Win=4193792 Len=0
351.30.216786	150.171.22.14	192.168.1.202	TCP	54	443 → 55072 [ACK] Seq=4996 Ack=2381 Win=4193792 Len=0
313.20.162882	151.101.112.217	192.168.1.202	TCP	54	443 → 61844 [ACK] Seq=1 Ack=74 Win=286 Len=0
314.20.162882	151.101.112.217	192.168.1.202	TCP	54	443 → 61844 [ACK] Seq=1 Ack=120 Win=286 Len=0
63.4.008136	170.76.137.89	192.168.1.202	TCP	54	443 → 51940 [ACK] Seq=46 Ack=88 Win=7925 Len=0
147.16.013631	170.76.137.89	192.168.1.202	TCP	66	443 → 63995 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 SACK_PERM
251.16.138066	170.76.137.89	192.168.1.202	TCP	54	443 → 63995 [ACK] Seq=1 Ack=1754 Win=5893 Len=0
258.16.257432	170.76.137.89	192.168.1.202	TCP	54	443 → 63995 [ACK] Seq=152 Ack=1005 Win=5944 Len=0
259.16.257432	170.76.137.89	192.168.1.202	TCP	54	[TCP Dup ACK 258#1] 443 → 63995 [ACK] Seq=152 Ack=1005 Win=5944 Len=0
260.16.268248	170.76.137.89	192.168.1.202	TCP	54	443 → 63995 [ACK] Seq=152 Ack=3910 Win=8049 Len=0
271.16.458079	170.76.137.89	192.168.1.202	TCP	66	443 → 58695 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 SACK_PERM

> Frame 247: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{16335CB7-C7...}

> Ethernet II, Src: WNC_72:09:00 (24:41:fe:72:09:00), Dst: CloudNetwork_7b:3c:bb (10:b1:df:7b:3c:bb)

> Internet Protocol Version 4, Src: 170.76.137.89, Dst: 192.168.1.202

▼ Transmission Control Protocol, Src Port: 443, Dst Port: 63995, Seq: 0, Ack: 1, Len: 0

Source Port: 443

Destination Port: 63995

[Stream index: 17]

[Stream Packet Number: 2]

> [Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 603049892

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

0000 10 b1 df 7b 3c bb 24 41 fe 72 09 00 08 00 45 00

0010 00 34 a5 35 40 00 fa 06 e5 75 aa 4c 89 59 c0 a8

0020 01 ca 01 bb f9 fb 23 f1 cf a4 94 a5 04 fd 80 12

0030 10 2c e1 25 00 00 02 04 05 64 01 03 03 00 04 02

0040 00 00

normal capture.pcapng

Packets: 6438 - Displayed: 1835 (28.5%) - Dropped: 0 (0.0%)

Profile: Default

12:42 06-11-2025